Using Access Control Lists to Modify File System Permissions on Windows Systems (3e)

Security Strategies in Windows Platforms and Applications, Third Edition - Lab 02

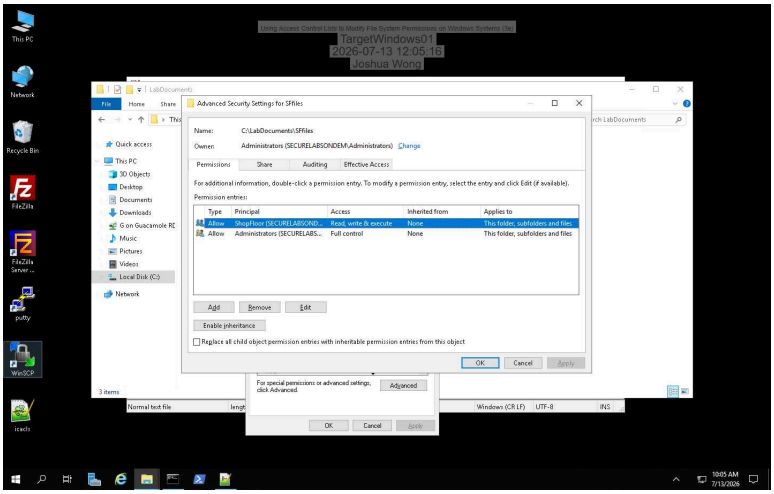
Student: Joshua Wong
Email: joshuawong7470@gmail.com
Report Generated: Monday, July 13, 2026 at 2:26 PM

Time on Task: 1 hour, 34 minutes
Progress: 100%

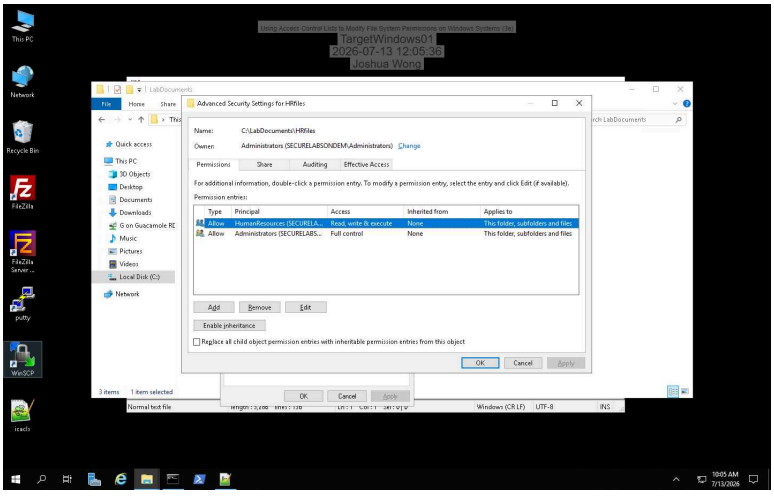
Section 1: Hands-On Demonstration

Part 1: View Existing ACLs on a Windows System

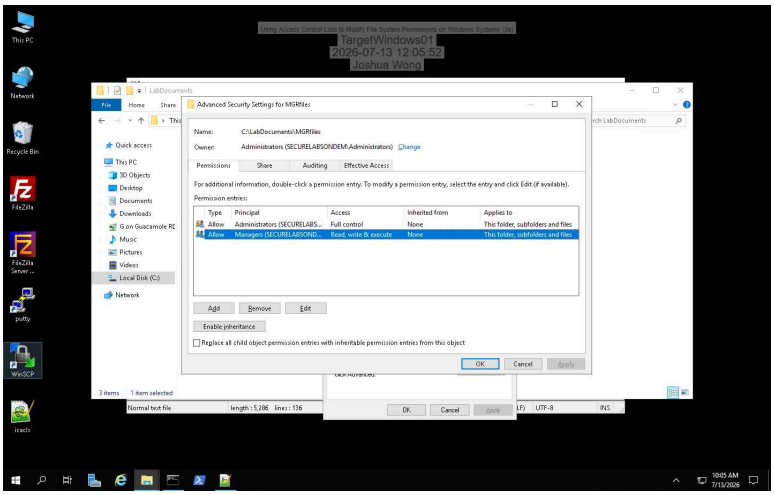
5. **Make a screen capture** showing the **current permissions for the SFfiles folder**.



12. **Make a screen capture** showing the **current permissions for the HRfiles folder.**



15. **Make a screen capture showing the current permissions for the MGRfiles folder.**



Part 2: Modify ACLs using Icacls.exe

8. **Compare the results** of the icacls.exe command with the ACLs you reviewed in Part 1 of this lab. Do they match?

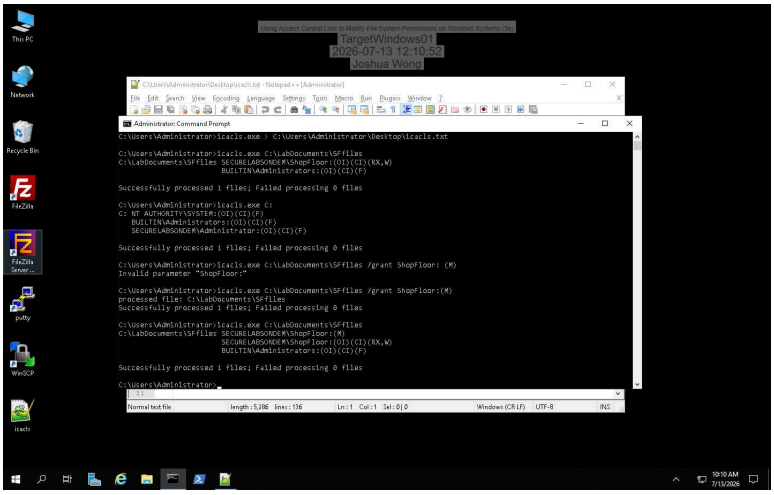
The permissions of the folder revealed through the icacls.exe command matches up with the permissions that showed up earlier for each user or group. Additionally the icacls.exe command shows the inheritance rights for each user or group in slightly different but perhaps more detailed terms.

11. **Compare the new results** of this command with the results from step 7.

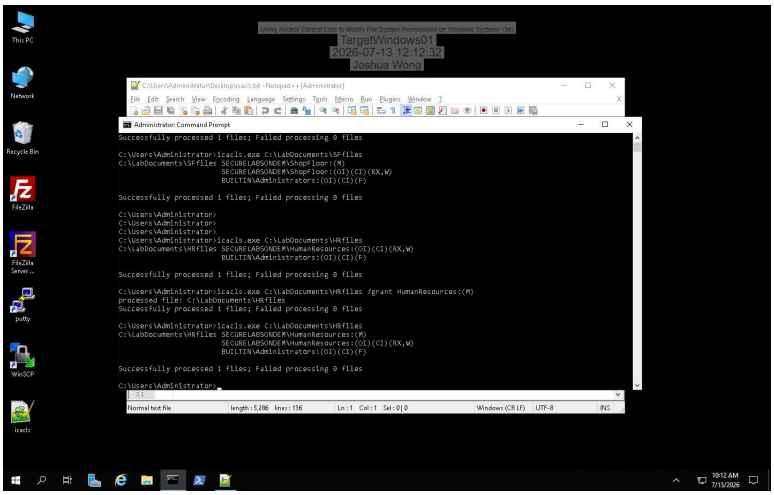
    You should notice that there is now a new line item for SECURELABSONDEM\ShopFloor. The grant command creates a new line item for any principal with permissions added through icacls.exe; technically, icacls.exe adds “special” permissions, which are visible by clicking the Advanced button in the folder Properties dialog box.

There is now a new line "SECURELABSONDEM\ShopFloor:(M) when SFfiles folder is enumerated.

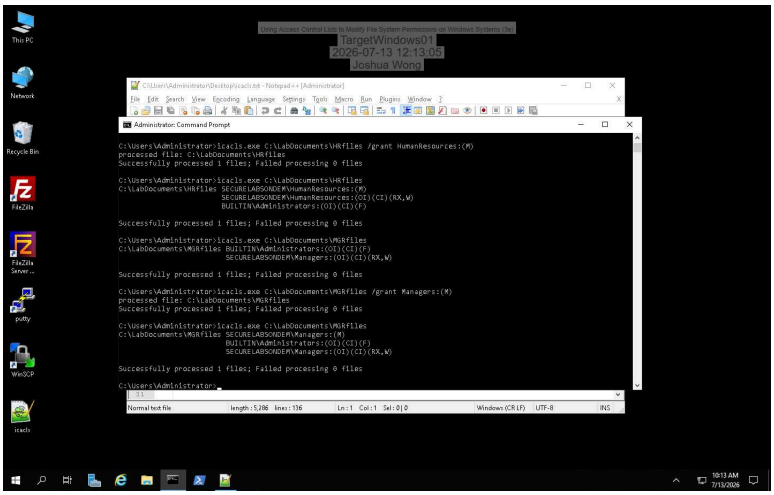
12. **Make a screen capture** showing the **changes you made to the SFfiles folder permissions**.



14. **Make a screen capture** showing the **changes you made to the HRfiles folder permissions**.



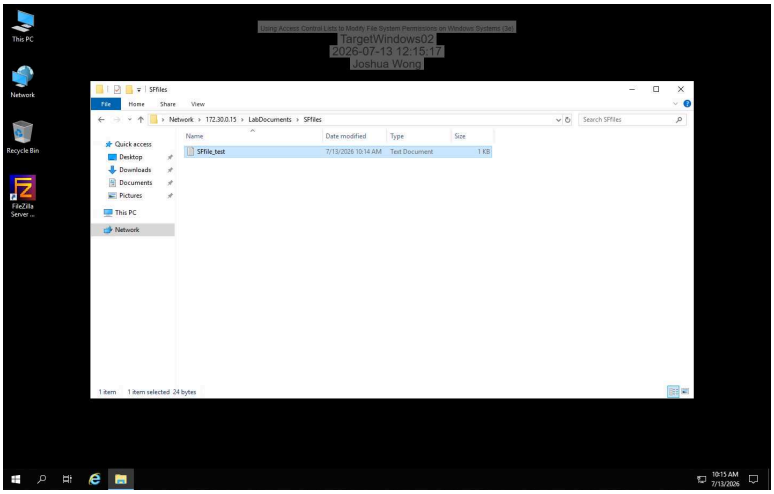
16. **Make a screen capture** showing the **changes you made to the MGRfiles folder permissions**.



Part 3: Validate ACL Settings

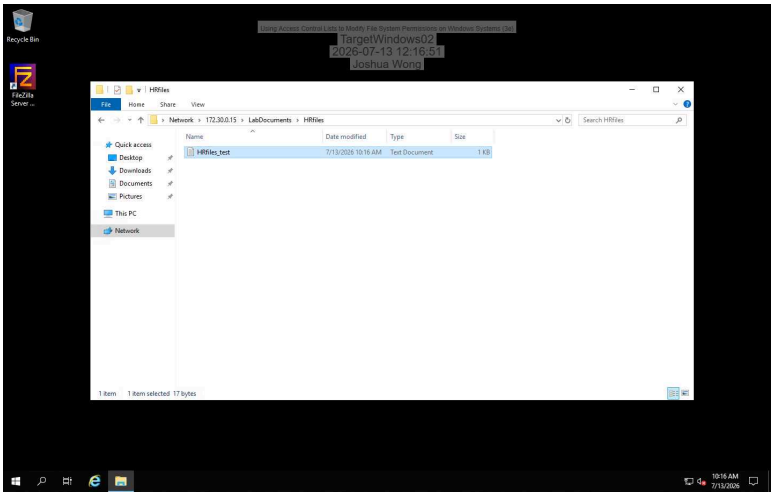
14. **Make a screen capture** showing the **modified text file in the SFfiles folder**.

    The modified file will show a 1 KB value in the Size column, indicating that text has been added to the file.



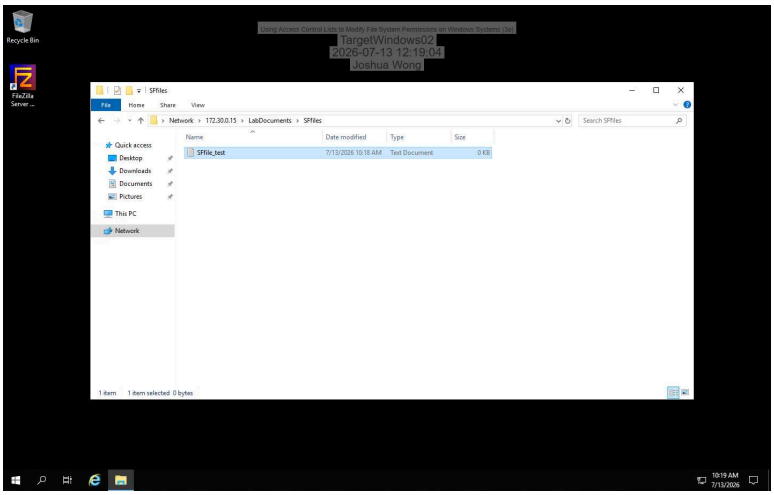
17. **Make a screen capture** showing the **modified text file in the HRfiles folder**.

    The modified file will show a 1 KB value in the Size column, indicating that text has been added to the file.



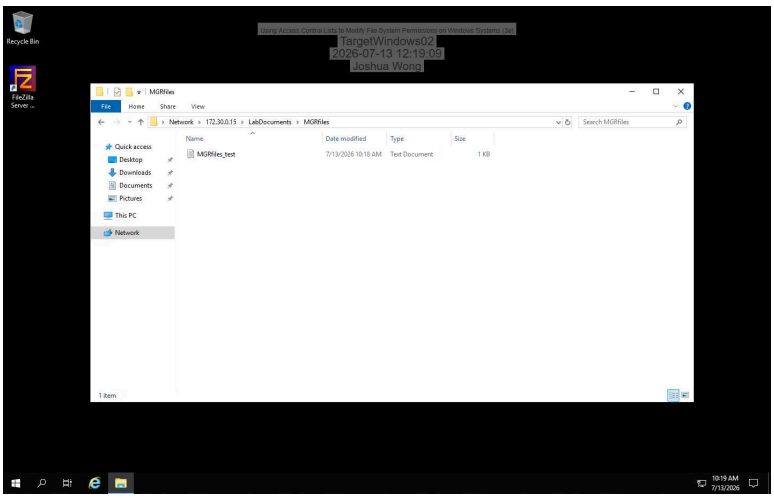
20. **Make a screen capture** showing the **re-modified text file in the SFfiles folder**.

    The modified file will show a 0 KB value in the Size column, indicating that all text has been removed from the file.



21. **Make a screen capture** showing the **modified text file in the MGRfiles folder**.

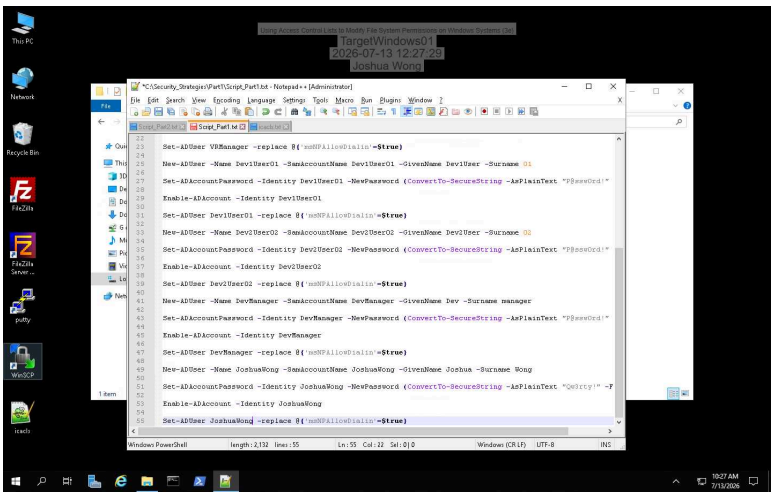
    The modified file will show a 1 KB value in the Size column, indicating that text has been added to the file.



Section 2: Applied Learning

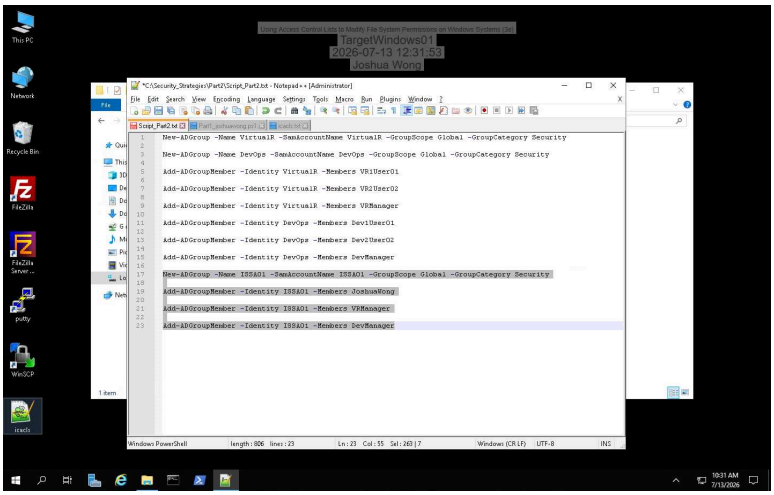
Part 1: Add a New User using a Script

3. **Make a screen capture** showing **the new user account in this script**.



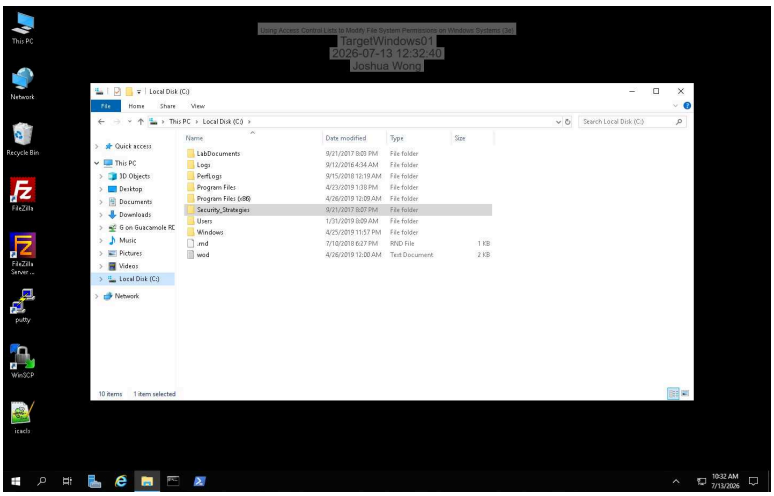
Part 2: Add a New Group using a Script

5. **Make a screen capture** showing the **modifications you made to the script**.



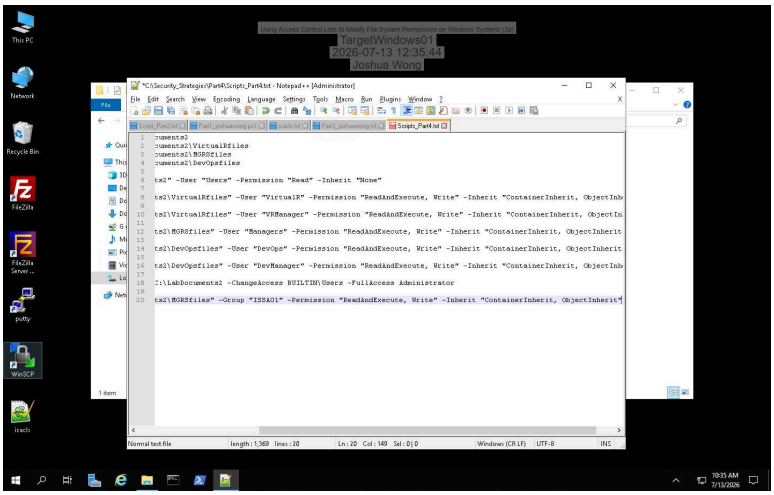
Part 3: Modify Permissions using a Script

2. **Make a screen capture** showing the **current contents of the TargetWindows01 C: drive**.

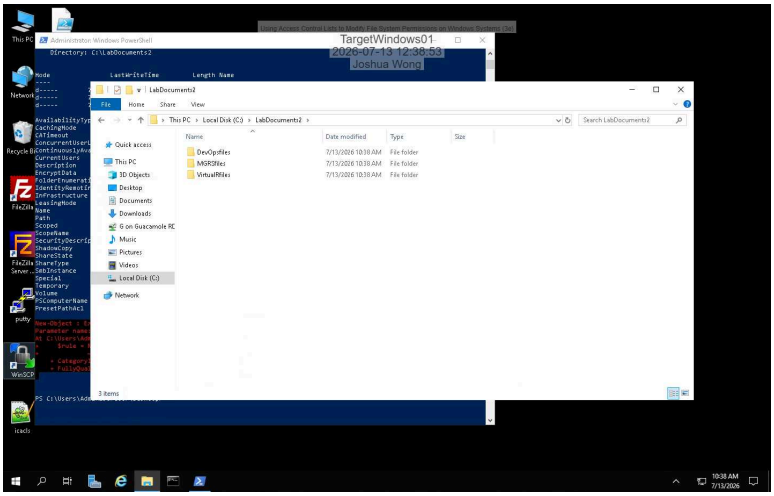


Part 4: Create Directories using a Script

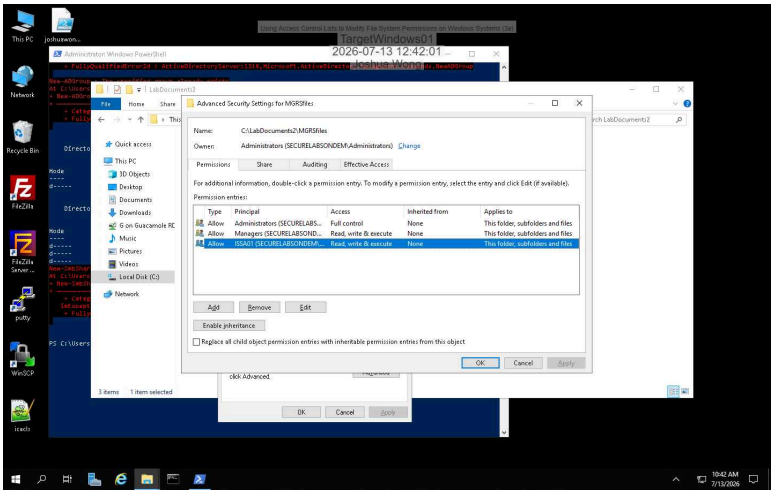
3. **Make a screen capture** showing the **modifications to the final part of the script**.



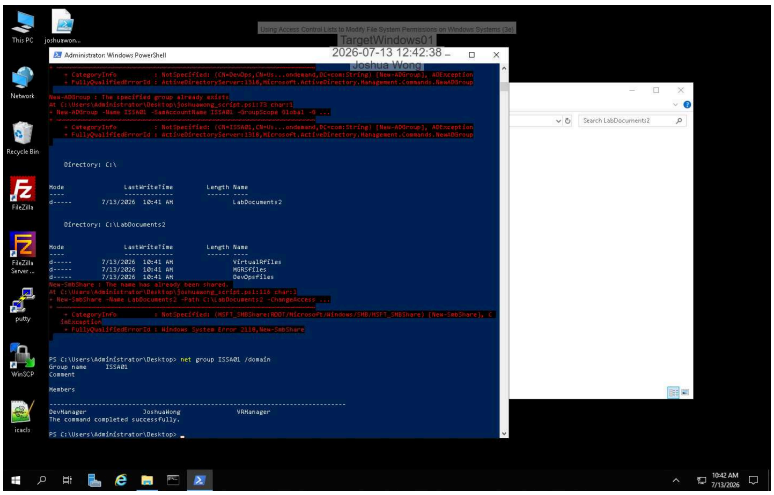
10. **Make a screen capture** showing the **contents of the new LabDocuments2 directory**.



12. **Make a screen capture** showing **the permissions for the ISSA01 security group**.



15. **Make a screen capture** showing the **members of the ISSA01 security group**.



Section 3: Challenge and Analysis

Part 1: Analysis and Discussion

Explain how the principle of least privilege can be used in a corporate setting to protect corporate resources.

Principle of least privilege is used to limit each individual's permissions and authorization to the least amount to get their task or job done. In a corporate setting this protects important resources such as classified files, financial information, and private customer data from being accessed by just anyone that signs onto the company; like a janitor, contractor, or a software developer if they do not need private or critical resources for their tasks. If this principle isn't applied, bad actors can sign on into the company as an insider to access critical information and use that for extortion or use their permission to escalate attacks. Another example is that bad actors can compromise a intern or a new hire's company account since they may be more vulnerable, and use their permissions to escalate or stage and attack. However, if permissions of each individual in the company are set accordingly it adds an additional layer of security across the organization.

Part 2: Tools and Commands

Research ACLs on the Internet and **identify** the permissions required to rename existing files.

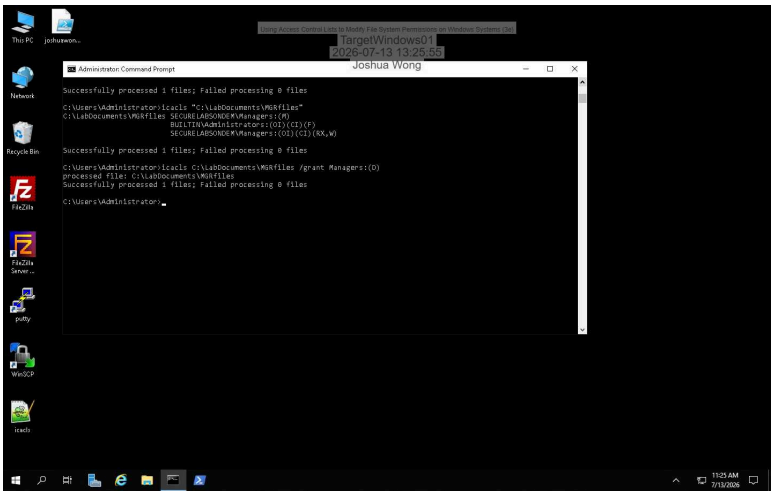
to rename existing files, it requires the permission to be able to execute in the directory to access the directory and its contents, read the directory in order to see the file to be renamed, and to write to the directory to rename the file.

Part 3: Challenge Exercise

Describe the choices you've made.

I used "icacls.exe C:\LabDocuments\MGRfiles" to first check the permissions that are granted to the managers group. Then I used "icacls.exe C:\LabDocuments\MGRfiles /grant Managers:(D)", which the cmd replied it successfully processed the files

Make a screen capture showing your **successfully executed icacls commands**.


Using Access Control Lists to Modify File System Permissions on Windows Systems Cont...
TargetWindows01
2026-07-13 13:25:55
Joshua Wong
Administrator: Command Prompt
Successfully processed 1 files; Failed processing 0 files
C:\Users\Administrator>icacls "C:\LabDocuments\MGRfiles"
C:\LabDocuments\MGRfiles SECURELABSONDEM\Managers:(M)
BUILTIN\Administrators:(OI)(CI)(F)
SECURELABSONDEM\Managers:(OI)(CI)(RX,W)
Successfully processed 1 files; Failed processing 0 files
C:\Users\Administrator>icacls C:\LabDocuments\MGRfiles /grant Managers:(D)
processed file: C:\LabDocuments\MGRfiles
Successfully processed 1 files; Failed processing 0 files
C:\Users\Administrator>
This PC
joshuawon...
Network
Recycle Bin
FileZilla
FileZilla Server ...
putty
WinSCP
icacls
11:25 AM
7/13/2026